ABSTRACT

The Public Distribution System (PDS) plays a vital role in ensuring food security, but existing systems face challenges such as inaccurate demand estimation, inefficient stock management, lack of real-time monitoring, and vulnerability to fraud. To address these issues, this project proposes an **AI-Driven Cloud-Based Smart T-Ration Analytics System** that transforms traditional ration distribution into a data-driven and intelligent system.

The proposed system leverages **cloud computing and data analytics** to store, process, and analyse large-scale ration distribution data efficiently. A centralized cloud database enables real-time data access, while analytical modules such as **demand forecasting, smart stock allocation, and real-time inventory tracking** improve decision-making and resource utilization.

In addition, the system incorporates advanced features such as **anomaly detection, shortage prediction, and geo-analytics visualization**, allowing authorities to identify high-risk regions and take proactive measures. These features are not widely implemented in current real-world systems and provide a significant improvement over traditional approaches.

The results demonstrate enhanced accuracy in demand prediction, improved stock allocation, and better transparency in the distribution process. Overall, the system provides a **scalable, efficient, and future-ready solution** for optimizing ration distribution and strengthening food security management.

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